NEWTON SCHOOL OF TECHNOLOGY
Data Visualization & Analytics — Capstone 2

Identifying High-Risk Road Conditions and Geographic Hotspots to Reduce Traffic Accident Severity Across the United States

Sector: Transportation / Public Safety

Team ID	B_G14
Section	DVA-B
Institute	Newton School of Technology
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GitHub Repository	github.com/AalokeCode/B_G14_DVACapstone
Tableau Dashboard	public.tableau.com — USARoadAccidentsIntelligence

Team Members

Name	GitHub	Role	Responsibility
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1. Executive Summary

Road traffic accidents impose an estimated $340 billion annual cost on the US economy. Despite large-scale data availability, resource allocation by transportation authorities remains predominantly reactive. This project closes that gap by mining seven years of geospatially indexed accident data to produce prioritised, evidence-based recommendations.

1.1 Problem

Which geographic locations, road conditions, and temporal patterns are most strongly associated with high-severity traffic accidents in the United States (2016–2023), and what targeted interventions can reduce accident severity at scale?

1.2 Approach

Analysed the Kaggle US Accidents dataset (7.7M records; sampled to 89,894 for depth) using a Python ETL pipeline across five Jupyter notebooks, six formal statistical tests, and an interactive Tableau dashboard published on Tableau Public.

1.3 Key Insights & Recommendations

Key Insights	Key Recommendations
• 21.3% of accidents are high-severity (Severity 3–4), driving disproportionate emergency cost. • Severity-4 blocks roads 2.9× longer than Severity-1 (130 min vs 44.8 min). • Traffic signals nearly halve high-severity odds (OR = 0.47). • Junctions (OR = 1.35) and railway crossings (OR = 1.44) are top risk features. • Weather significantly determines severity (χ^2 = 3,527.80, $p \approx 0$).	1. Redesign top-50 high-severity junctions — est. 10–15% severity reduction. 2. Expand signal deployment at uncontrolled intersections — est. 7–8% reduction/site. 3. Upgrade railway crossings with active warning systems. 4. Deploy weather-responsive variable speed limits on high-risk corridors. 5. Implement severity-tiered emergency staging — est. 20–30 min faster clearance.

2. Sector & Business Context

2.1 Sector & Decision-Maker

The Transportation and Public Safety sector encompasses state DOTs, federal highway agencies, and emergency services. US road infrastructure investment exceeds $200 billion annually, yet accident rates have not declined proportionately. The primary decision-maker is the State DOT Director controlling road safety budgets, emergency response staffing, and infrastructure improvement programmes. Secondary stakeholders include municipal traffic engineers and federal NHTSA analysts.

2.2 Business Value

A 10% reduction in Severity-4 accident frequency on the top-100 high-risk corridors saves an estimated $340 million annually. Every 10-minute reduction in Severity-4 clearance time on a high-volume interstate saves $500K–$1M in productivity loss. The analysis delivers an evidence-ranked priority list enabling DOTs to maximise safety ROI from constrained budgets.

3. Problem Statement & Objectives

3.1 Formal Definition

> *Which geographic locations, road conditions (infrastructure, weather), and temporal patterns are most strongly associated with high-severity traffic accidents in the United States from 2016 to 2023, and what targeted interventions would reduce accident severity at scale?*

3.2 Scope & Success Criteria

In Scope	Out of Scope
• Severity analysis (levels 1–4) on Kaggle US Accidents dataset • Geographic hotspot identification (state & city) • Temporal analysis (hour, day, month, season, YoY) • Infrastructure POI analysis (junctions, signals, railway, crossings) • Weather-severity association testing • Logistic regression to quantify high-severity odds • Interactive Tableau dashboard with 4+ filter dimensions	• Driver behaviour or vehicle type data • Population / VMT normalisation • Real-time predictive scoring in production • Fatality or injury data (severity = traffic impact)

Success Criteria: ≥5 statistically significant findings (α=0.05); logistic regression with ≥3 actionable predictors at p<0.001; Tableau dashboard published publicly; 3–5 recommendations with quantified expected impact; all notebooks on GitHub with reproducible outputs.

4. Data Description

Attribute	Details
Dataset	US Accidents (Feb 2016 – Mar 2023) — Moosavi et al., Kaggle
Raw Records	7,728,394 rows × 46 columns
Working Sample	89,894 rows × 48 columns (post-feature engineering)
Coverage	49 US states | February 2016 – March 2023 | CSV format
Target Variable	severity (int 1–4): 1=low, 4=critical traffic impact

Column	Type	Description	Role
severity	int 1–4	Traffic impact (1=low, 4=critical)	Primary target
start_time	datetime	Accident start timestamp	Temporal features
start_lat/lng	float	GPS coordinates	Geographic mapping
weather_condition	string	Text weather label	Weather analysis
duration_min	float	Minutes road blocked (derived)	Response load KPI
junction / traffic_signal / railway / crossing	bool	Infrastructure feature within ~150 m	Infrastructure analysis
is_high_severity	bool	Severity ≥ 3 (derived)	Binary regression target

4.1 Data Limitations

- Geographic over-representation: CA, FL, TX dominate counts — severity rates used for cross-state comparisons.

- Weather station proximity: values from nearest airport, potentially miles from accident site.
- Severity assignment noise: 1–4 score set by third-party API at incident time, not post-hoc verified.
- 2021 spike: post-2020 volume rise partly reflects API coverage expansion, not a genuine accident increase.

5. Data Cleaning & ETL Pipeline

Pipeline committed across notebooks/01_extraction.ipynb and notebooks/02_cleaning.ipynb; modular version in scripts/etl_pipeline.py.

Column(s)	Missing Rate	Treatment
end_lat, end_lng	~50%	Dropped — redundant with start coordinates
number (street no.)	~60%	Dropped
severity, start_time, start_lat/lng	<1%	Rows dropped (target & key features cannot be imputed)
Numeric weather fields	5–15%	Median imputation
String geography fields	<5%	Filled with 'Unknown'
Boolean POI flags	Minimal	Filled with False

5.1 Outlier Treatment & Feature Engineering

Feature	Logic	Business Meaning
duration_min	(end_time − start_time) in minutes; negative removed, >99th pct capped	Emergency response load
is_rush_hour	hour ∈ [7,9] ∪ [16,19]	Peak staffing windows
is_weekend	day_of_week ≥ 5	Weekend patrol planning
season	Dec–Feb=Winter, Mar–May=Spring, Jun–Aug=Summer, Sep–Nov=Fall	Seasonal budget planning
is_high_severity	severity ≥ 3	Binary logistic regression target
temperature_f	Values outside [−60, 140]°F removed as physically impossible	Weather analysis
distance_mi	Capped at 99th percentile	Road impact metric

6. KPI & Metric Framework

KPI	Formula	Value	Notebook
High-Severity Rate	is_high_severity.mean() × 100	**21.3%**	nb04
Median Clearance — Sev-4	groupby('severity')['duration_min'].median()	**130 min**	nb04
Median Clearance — Sev-1	groupby('severity')['duration_min'].median()	**44.8 min**	nb04
Hotspot Concentration	groupby('state')['is_high_severity'].agg(['count','mean'])	CA, FL, TX lead	nb03
Adverse-Weather Sev. Rate	Chi-square test + conditional rate by weather	**χ^2=3,527.80**	nb04

7. Exploratory Data Analysis

7.1 Severity Distribution

Severity 2 dominates at 77.7% of all accidents. High-severity events (Severity 3+4) represent only 21.3% but impose disproportionate costs — Severity-4 median clearance is 130 min vs 44.8 min for Severity-1. Emergency budgets should therefore be sized against the 21.3% share, not the modal Severity-2 class; a 1% reduction in high-severity events delivers more economic value than a 5% reduction in Severity-2.

7.2 Geographic Hotspot Analysis

California, Florida, and Texas lead raw accident counts. However, the high-severity rate (proportion of accidents that are Severity 3+4) varies independently of volume. Budget allocation should follow severity rate, not absolute count — a state with moderate volume but elevated severity rate delivers higher per-dollar ROI. This distinction is central to the hotspot prioritisation recommended in Section 11.

7.3 Temporal Patterns

Dimension	Key Finding
Hour of Day	Volume peaks 7–9 AM and 4–7 PM. Nighttime severity is slightly elevated (mean 2.238 night vs 2.227 day), pointing to a lighting deficit.
Day of Week	Weekend accidents carry 19% higher odds of high-severity (OR = 1.19, $p < 0.001$) due to higher-speed, off-peak driving behaviour.
Month / Season	Winter months (Dec–Feb) produce both higher volume and a higher proportion of severe outcomes. Recurring December–January spike confirmed by seasonal decomposition.
Year-over-Year	Upward volume trend 2016–2021. Sharp 2021 spike partly attributable to API coverage expansion — flagged as a limitation.

7.4 Infrastructure POI Correlations

POI	Point-Biserial r	p-value	Direction
Crossing	−0.125	≈ 0	Protective
Traffic Signal	−0.124	≈ 0	Protective
Stop	−0.057	≈ 0	Protective
Junction	+0.056	≈ 0	Risk Factor
Railway	−0.011	0.001	Logistic regression reverses this (see Sec. 8.6)

7.5 Weather Analysis

Clear weather accounts for the majority of accident volume, but adverse conditions (snow, heavy rain, fog) produce a materially different severity distribution. The chi-square test (Section 8.1) confirms this association is not due to chance.

8. Statistical Analysis

All tests implemented in notebooks/04_statistical_analysis.ipynb using scipy and statsmodels. Significance threshold α = 0.05.

8.1 Chi-Square — Severity Independence from Weather

Test	Result	Business Interpretation
H_0: Severity and weather are independent. H_1: They are associated.	**χ^2 = 3,527.80, df = 27** p ≈ 0, Cramér's V = 0.119 **Decision: Reject H_0**	Adverse conditions produce systematically more severe accidents. Weather-responsive variable speed limits and dynamic message signs are statistically justified on high-volume corridors.

8.2 Kruskal–Wallis — Duration Across Severity Levels

Non-parametric test used due to heavy right skew in duration data. H = 6,597.67, p ≈ 0. Reject H_0 — duration distributions differ significantly across severity levels.

Severity 1	Severity 2	Severity 3	Severity 4
44.8 min	75.7 min	44.4 min	**130.0 min**

Severity-4 blocks traffic 2.9× longer than Severity-1. Routing emergency teams by predicted severity — not just volume — can reclaim thousands of person-hours annually.

8.3 Mann–Whitney U — Traffic Signals and Severity

H_0: Severity distributions identical with/without a signal. U = 448,844,871, p ≈ 0. Reject H_0. Mean severity at signal-present locations = 2.086 vs 2.257 absent — a 7.6% reduction. Expanding signal deployment at uncontrolled high-volume intersections is the highest-evidence, most scalable intervention in the dataset.

8.4 Spearman Correlation — Numeric Features vs Severity

Feature	Spearman ρ	p-value
duration_min	**−0.179**	≈ 0
distance_mi	−0.117	≈ 0
pressure_in	+0.086	≈ 0
wind_speed_mph	+0.051	≈ 0
precipitation_in	+0.028	1.2×10^{-16}

Duration and distance are the top numeric correlates with severity, justifying severity-scaled emergency pre-positioning.

8.5 Time-Series Decomposition

Monthly accident counts (Feb 2016 – Mar 2023) decomposed using an additive model with 12-month period. Clear seasonal component with recurring December–January peaks and summer troughs. Upward trend 2016–2021. Winter staffing plans, de-icing operations, and response pre-positioning should align with this predictable seasonal window.

8.6 Logistic Regression — Odds of High-Severity Outcome

Target: is_high_severity (severity ≥ 3). Predictors: weather numerics + infrastructure booleans + temporal flags. 89,894 observations. Pseudo R^2 = 0.036; LLR p ≈ 0.

Predictor	Coefficient	Odds Ratio	p-value
Railway	0.366	**1.44 ▲**	0.0003

Junction	0.302	**1.35 ▲**	≈ 0
Is Weekend	0.175	1.19 ▲	≈ 0
Is Rush Hour	0.063	1.07 ▲	0.0002
Traffic Signal	−0.748	**0.47 ▼**	≈ 0
Crossing	−1.007	0.37 ▼	≈ 0
Stop	−1.382	0.25 ▼	≈ 0

Railway crossings (OR = 1.44) and junctions (OR = 1.35) are the highest-risk features. Traffic signals (OR = 0.47) are the strongest single protective intervention in the dataset — nearly halving high-severity odds.

9. Tableau Dashboard Design

Dashboard URL: public.tableau.com/app/profile/somraj.nandi/viz/USARoadAccidentsIntelligence/Dashboard1

Dashboard 1 — Executive View	Dashboard 2 — Operational View
Target user: State DOT Director. Goal: full national KPI status in under 60 seconds. • National KPI tiles: total accident count, 21.3% high-severity rate, top-5 states by volume • Severity distribution donut chart (Severity 1/2/3/4 share) • Year-over-year trend line (2016–2021 trajectory) • Geographic choropleth / point map of accident density by state	Target user: Traffic engineer. Goal: quarterly infrastructure investment planning. • City-level severity-coded hotspot map (drill down: state → city) • Temporal heatmap: hour-of-day × day-of-week accident density • Weather condition bar chart: severity by condition • Infrastructure POI comparison: junction vs signal vs crossing severity

9.1 Interactive Filters

Filter	Filter	Filter	Filter
State selector	Severity level (1–4)	Weather condition	Year range (2016–2023)

10. Key Insights

Ten decision-oriented findings for transportation planners and public safety directors.

#	Insight
1	21.3% of accidents are high-severity but drive the majority of emergency cost. Severity 2 is 77.7% of events, yet Severity 3+4 demands disproportionate resources. Any 1% reduction in high-severity delivers more value than 5% reduction in Severity-2.
2	Severity-4 blocks traffic 2.9× longer than Severity-1 (130 min vs 44.8 min). Emergency teams routed by predicted severity can reclaim thousands of person-hours of road capacity annually.
3	Junctions are the highest-volume, high-severity infrastructure type (OR = 1.35). Accidents near junctions are 35% more likely to be high-severity. Junction redesign (roundabouts, raised crosswalks) offers the highest structural ROI.
4	Traffic signals nearly halve high-severity odds (OR = 0.47). Signal-present locations average severity 2.086 vs 2.257 absent — a 7.6% reduction. This is the strongest single protective intervention in the dataset, scalable to thousands of uncontrolled intersections.
5	Railway crossings carry the highest per-event severity uplift (OR = 1.44). Despite low volume relative to junctions, railway-proximate accidents are 44% more likely to be high-severity — most involve unprotected or poorly lit crossings.
6	Weather condition significantly determines severity (χ^2 = 3,527.80, $p \approx 0$, Cramér's V = 0.119). Snow, heavy rain, and fog produce materially worse severity distributions — weather-responsive speed limits are statistically justified.
7	Weekend accidents are 19% more likely to be high-severity (OR = 1.19, $p < 0.001$). Higher-speed, off-peak driving shifts the severity profile upward — weekend-specific patrol deployment in high-severity corridors is warranted.
8	Nighttime accidents are measurably more severe (mean 2.238 vs 2.227 daytime). Across millions of annual events this translates to thousands of additional high-severity incidents attributable to lighting deficit.
9	California, Florida, and Texas dominate volume, but severity rates vary independently. Budget allocation should follow high-severity rate per state, not absolute count, to maximise safety ROI per dollar spent.
10	Winter months produce recurring peaks in both volume and severity. Seasonal decomposition confirms a December–January spike. Staffing plans, de-icing operations, and response pre-positioning should align with this predictable annual window.

11. Recommendations

#	Insight Driver	Recommendation	Feasibility	Expected Impact
1	Junction OR = 1.35; highest-volume high-severity POI	Redesign top-50 high-frequency, high-severity intersections with roundabouts and raised crosswalks — identified in the Tableau hotspot map.	Medium — municipal planning approval; $500K–$2M per site.	10–15% severe junction accident reduction; $2M+ savings per redesigned corridor.
2	Signal OR = 0.47; strongest single protective intervention	Expand traffic signal deployment at uncontrolled high-volume intersections in top-10 hotspot cities.	Low–Medium — off-the-shelf hardware; primary barrier is permitting.	7–8% severity reduction per site; scalable at $50K–$200K per signal.
3	Railway OR = 1.44; highest per-event severity uplift	Upgrade all unsignalised railway crossings in high-accident corridors with active warning systems and improved lighting.	Medium — FRA coordination required.	15–20% reduction in crossing-adjacent high-severity events.
4	Weather χ^2 = 3,527.80, $p \approx 0$	Deploy weather-responsive variable speed limits and real-time advisory signs on high-volume corridors with frequent adverse conditions.	Medium — sensor and sign capital; aligns with MUTCD guidance.	Reduces adverse-weather severity premium; justified at $\alpha = 0.001$.
5	Sev-4 median clearance = 130 min	Implement severity-tiered emergency staging: pre-position heavy-response units in Sev-3/4-dominant corridors during peak risk windows (rush hour, winter weekends).	Low — operational change, zero capital outlay.	20–30 min faster Sev-4 clearance; reduces secondary incident risk.

12. Impact Estimation & Cost of Inaction

Lever	Estimated Impact	Logic
Signal expansion — top-200 uncontrolled intersections	~7.6% severity reduction/site; up to 15,000 fewer high-severity events/year nationally	OR = 0.47 applied to 21.3% high-severity base rate at signal-absent locations
Junction redesign — top-50 corridors	10–15% local severity reduction; $2M+ economic savings per corridor	OR = 1.35 reversal through redesign; $500K–$1M productivity/10 min clearance reduction
Severity-tiered response staging	20–30 min faster Severity-4 clearance; reduces secondary accident risk	Sev-4 median 130 min; 15–23% clearance reduction through pre-positioning
Weather-responsive VMS — 50 corridors	5–10% adverse-weather severity reduction	Cramér's V = 0.119 (moderate effect); conservative 50% attenuation through compliance

> Cost of Inaction: Every 10-minute increase in Severity-4 clearance on a high-volume interstate costs an estimated $500K–$1M in productivity. Without junction redesign, the highest-severity POI class continues receiving only general-purpose funding. Without weather-responsive controls, adverse-weather periods produce statistically distinguishable excess severe events at $\alpha = 0.001$.

13. Limitations

Limitation	Detail & Impact
Working Sample	89,894 rows from 7.7M raw records. Patterns are statistically robust but geographic coverage may under-represent lower-accident states. Absolute state counts should not be extrapolated without rerunning on the full dataset.
Severity Score Reliability	The 1–4 severity score is assigned by a third-party traffic API at incident time measuring road blockage, not injury severity. Post-hoc verification is unavailable; classification noise exists.
No Injury/Fatality Data	Recommendations optimise for traffic impact reduction. Human casualty minimisation may require a supplementary objective function and integration of NHTSA FARS data.
Low Model R^2	Logistic pseudo-R^2 = 0.036 — the model explains 3.6% of severity variance. Key unobserved factors include driver behaviour, vehicle type, road geometry, speed limit, and enforcement levels.
No VMT Normalisation	Dataset contains no population or vehicle-miles-travelled data. Cross-state absolute count comparisons are not population-normalised; severity rates are used instead but share the same limitation.
Weather Station Proximity	Weather values joined from nearest airport station, potentially miles from the accident site. Micro-weather conditions at the exact location are not captured.
2021 Data Artefact	Post-2020 volume spike is partly an artefact of API coverage expansion, not a genuine accident increase. Year-over-year trend conclusions should be interpreted with this caveat.

14. Future Scope

Initiative	Description
Full-Dataset Replication	Rerun pipeline on all 7.7M rows to achieve state-level statistical power and enable robust population-normalised comparisons.
Fatality Data Integration	Incorporate NHTSA FARS injury and fatality records to align severity scoring with human outcome metrics rather than road blockage duration.
Road Geometry Features	Add lane count, speed limit, curve radius, and pavement condition as predictors — likely to raise logistic pseudo-R^2 substantially beyond 0.036.
Real-Time Risk Scoring	Live API ingesting current weather, time of day, and road features to output a severity risk score for any GPS coordinate — enabling dynamic routing advisories and pre-incident staging.
Policy Simulation Tool	Interactive tool using logistic OR estimates to project accident reduction for any proposed infrastructure investment (e.g., 'install signals at these 50 intersections → predicted Δ high-severity rate').
VMT Normalisation	Integrate FHWA vehicle-miles-travelled data at state level to produce risk-per-VMT metrics enabling fair cross-state comparisons independent of population and driving exposure.

15. Conclusion

The United States road accident problem is large, costly, and — critically — predictable. Using 89,894 records from the Kaggle US Accidents dataset (February 2016 – March 2023), this project applied a structured Python ETL pipeline, six formal statistical tests, and an interactive Tableau dashboard to transform raw accident data into actionable intelligence for state DOTs and public safety directors.

The analysis establishes that high-severity accidents are not randomly distributed: they concentrate around junctions (OR = 1.35) and railway crossings (OR = 1.44), are significantly more frequent on weekends and in adverse weather (χ^2 = 3,527.80, $p \approx 0$), and are strongly mitigated by traffic signals (OR = 0.47). Severity-4 events block traffic for a median of 130 minutes — nearly three times the Severity-1 median — imposing disproportionate economic cost for their 21.3% frequency share.

The five recommended actions — junction redesign, signal expansion, railway crossing upgrades, weather-responsive controls, and severity-tiered emergency staging — are each directly supported by statistically significant findings and are feasible within existing DOT capital and operational frameworks. Executing all five on identified high-priority corridors has the potential to prevent thousands of high-severity events annually and reclaim millions of hours of road capacity across the United States.

Statistical Finding	Infrastructure Insight	Operational Insight
• χ^2 = 3,527.80 weather-severity link • H = 6,597.67 clearance-severity link • U = 448M signal protective effect • Pseudo-R^2 = 0.036 model baseline	• Railway crossings: OR = 1.44 (highest risk) • Junctions: OR = 1.35 (highest volume risk) • Traffic signals: OR = 0.47 (best protection) • Top-50 junctions identified in Tableau	• Severity-4 = 130 min clearance target • Winter peaks → seasonal pre-positioning • Weekend patrol uplift → OR 1.19 • 5 recommendations, all actionable now

Appendix A — Statistical Findings Summary

#	Test	Finding	Recommendation Mapped
1	Chi-Square — Weather × Severity	χ^2 = 3,527.80, p ≈ 0, Cramér's V = 0.119	Deploy weather-responsive variable speed limits and VMS on high-risk corridors
2	Kruskal–Wallis — Duration × Severity	H = 6,597.67, p ≈ 0; Sev-4 = 130 min vs Sev-1 = 44.8 min	Severity-tiered emergency response staging; heavy-unit pre-positioning
3	Mann–Whitney U — Signal Presence	U = 448,844,871, p ≈ 0; 7.6% mean severity reduction	Expand signal deployment at uncontrolled high-volume intersections
4	Spearman Correlation — Numerics	duration_min ρ = −0.179; distance_mi ρ = −0.117	Severity-scaled emergency pre-positioning; road impact metrics as dispatch input
5	Time-Series Decomposition	Recurring Dec–Jan winter peaks; upward trend 2016–2021	Seasonal staffing and de-icing budget cycles aligned to winter peak
6	Logistic Regression — OR Table	Junction OR=1.35; Railway OR=1.44; Signal OR=0.47	Junction redesign + signal expansion + railway crossing upgrades

Appendix B — Data Dictionary (Key Derived Columns)

Column	Type	Logic	Business Use
duration_min	float	(end_time − start_time) in minutes; negatives removed; >99th pct capped	Emergency response load KPI
is_high_severity	bool/int	severity ≥ 3	Binary logistic regression target
is_rush_hour	bool	hour ∈ {7,8,9} ∪ {16,17,18,19}	Peak staffing window flag
is_weekend	bool	day_of_week ≥ 5 (Sat/Sun)	Weekend patrol planning
season	string	Dec–Feb=Winter, Mar–May=Spring, Jun–Aug=Summer, Sep–Nov=Fall	Seasonal budget cycles
severity_label	string	1→Low, 2→Moderate, 3→High, 4→Critical	Human-readable Tableau labels

Appendix C — Contribution Matrix

Team Member	Dataset	ETL	EDA	Stats	Tableau	Report	PPT/Viva
Aaloke Eppalapalli (AalokeCode)	Lead	Sup.	Sup.	Sup.	Sup.	Sup.	Lead
Aryan Kinha (aryankinha)	Sup.	Lead	Sup.	Sup.	Sup.	Sup.	Sup.
Punit Yadav Rao (punityadavrao)	Sup.	Sup.	Lead	Sup.	Sup.	Sup.	Sup.
Somraj Nandi (somraj112)	Sup.	Sup.	Sup.	Sup.	Lead	Sup.	Sup.
Alok (bigXalok)	Sup.	Sup.	Sup.	Sup.	Sup.	Lead	Sup.
Balajee	Sup.	Sup.	Sup.	Lead	Sup.	Sup.	Sup.

GitHub Contribution Evidence:

- aryankinha — PR #1, PR #2: full analytics pipeline (notebooks 01–05, ETL, data dictionary)
- somraj112 — PR #3: Tableau workbook and dashboard screenshots
- bigXalok — Report writing: drafted and structured the full project report

Contribution verifiable at: github.com/AalokeCode/B_G14_DVACapstone/graphs/contributors

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.